🧲 Section: Our Dynamic Universe

🔷 Topic: Collisions, Explosions and Impulse

Question 3 – Newton's Cradle

✔ 3 marks

A **Newton's Cradle** illustrates key principles in dynamics and momentum:

✅ Conservation of Momentum

- In the absence of external forces, total momentum is conserved.
- When one ball strikes the next, its momentum is transferred through the stationary balls to the one at the end.

✅ Conservation of Energy

- In ideal conditions (no energy lost to sound or heat), kinetic energy is conserved.
- The outgoing ball swings to a height similar to that of the incoming one.

✅ Elastic Collisions

- Collisions between the steel balls are nearly elastic.
- Little kinetic energy is lost, allowing energy and momentum to be transferred cleanly.

✅ Impulse (Optional Point)

- The change in momentum of each ball occurs over a short time.
- Force × time (impulse) governs the transfer of momentum between balls.

✏️ Sample Full Answer (3/3):

> A Newton's Cradle demonstrates conservation of momentum, as momentum is passed through the stationary balls to the one at the end.
> It also shows conservation of kinetic energy in an almost elastic collision.
> As a result, the end ball swings upward to almost the same height as the first ball was released.

🧠 Revision Tips

- Be ready to **explain physical laws**, not just name them.
- When asked to **comment on a demonstration**, state:
  1. The principle (e.g. momentum)
  2. The physics relationship (e.g. $p = mv$)
  3. What the demonstration shows (e.g. ball at other end moves off)
